

Real-World Order of Operations

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

Target I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

MTR.7.1 — Notes I-III, both Examples and boards 2, 3, 4, 6, 7 and 9 ·

MTR.2.1 — Notes II and board 9 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	12 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	12 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. Every substituted value goes into parentheses. That one habit prevents half of what goes wrong today.
2. There is no word that tells you which operation to use. Read what is happening, not which words are there.
3. Round once, at the very end, and write the unit. An answer with no unit is not an answer.

Slide by Slide

1. **Real-World Order of Operations.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite and it carries the day's one habit.
3. **Warm-Up — Answers.** Reveal. 1. 0 2. $25/2$ 3. $4/3$ 4. 18
4. **Notes — Copy both lines.** Every substituted value goes into parentheses. That single habit prevents half of what goes wrong in this lesson.
5. **Notes — Copy both problems.** There is no word that tells you which operation to use. Read what is happening, not which words are present.
6. **Notes.** Round once, at the very end. Rounding in the middle moves the answer, and it moves it further every step you take afterwards.
7. **Example 1 — A formula with a power in it.** SPOKEN ONLY.
8. **Example 1 — Worked.** Substitute in parentheses, square the speed, then multiply.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Same formula, smaller numbers.
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — A cost with a rate and a flat fee.** SPOKEN ONLY.
12. **Example 2 — Worked.** The two per-square-foot prices go inside the group that is multiplied by the area; the flat fee stays outside it.
13. **Your Turn — Same steps, your numbers. Boards up when done.** One rate and one flat fee this time.
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** Evaluate $3x^2$ when $x = -2$. → **12** *-12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to $3x$ [8.NSO.1.7: the exponent belongs to its own base]*
- 17-18. **Board 2.** A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v . Find her kinetic energy. → **2,240 joules** *The price of the bicycle is not needed. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as $2v$ [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules*
- 19-20. **Board 3.** A plan costs 15 dollars plus 4 dollars for each gigabyte. Find the cost of 6 gigabytes. → **39 dollars** *114 — $(15 + 4)(6)$: the flat 15 was charged six times [8.NSO.1.7: only the per-gigabyte price is repeated]; 25 — $15 + 4 + 6$, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]*
- 21-22. **Board 4.** A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost. → **284 dollars** *4,064 — $64(3.50 + 60)$: the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — $8(3.50) + 60$: the side used instead of the area*
- 23-24. **Board 5.** Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$. Which is the value? → **A — 180** *(B) v^2 was read as $2v$ [8.NSO.1.3: an exponent is not a factor]; (C) the one half was*

dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, $(\frac{1}{2} \cdot 40 \cdot 3)^2$ [8.NSO.1.7: the exponent belongs to the v alone] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

25-26. Board 6. A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost? → **$D - 12^2(1.25) + 38$** (A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. A fence runs along 6 sides, each $\sqrt{200}$ feet long. Find its total length, to the nearest hundredth of a foot. → **84.85 feet** 84 — the side was rounded to 14 before multiplying [Unit 2: round once, at the end]; 84.85 with no unit — the answer is in feet

29-30. Board 8. Evaluate $(a^2 - b) \div c$ when $a = 7$, $b = 4$, $c = 5$. → **9** 48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — $49 \div 5$ done first, then 4 subtracted

31-32. Board 9. A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes. Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax. Find both totals, and say why the key words could not have told you. → **$39 \text{ dollars and } 66 \text{ dollars}$** 39 and 66 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 39 twice — the second structure was not read

33. Independent Set — Six questions. On your own, in silence. Hand out the Independent Set.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. -12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]

Board 2. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]

Board 3. $114 - (15 + 4)(6)$: the flat 15 was charged six times [8.NSO.1.7: only the per-gigabyte price is repeated]

Board 4. $4,064 - 64(3.50 + 60)$: the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]

Board 5. v^2 was read as $2v$ [8.NSO.1.3: an exponent is not a factor]

Board 6. the side was used where the area is needed [8.NSO.1.7: read what the rate is per]

Board 7. 84 — the side was rounded to 14 before multiplying [Unit 2: round once, at the end]

Board 8. 48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]

Board 9. 39 and 66 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]

Materials Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.